



RS6xxx Production Testing Guide

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1 Preface

1.1 Purpose

This document primarily introduces how to establish a testing engineering process for the RS6xxx chip solution during assembly production.

The document includes test case descriptions, development guidelines, and instructions for setting up the testing environment.

Users can refer to [Section 2.3 "Production Testing System Development Guide"](#) for quick start.

1.2 Target Audience

This document is provided for test engineers, hardware development engineers, test software development engineers, and structural engineers in the factory.

2 Development Guide

2.1 Introduction To RS6xxx

RS6xxx is a 60 GHz FMCW MIMO radar sensor chip from Possumic Technology. The chip integrates mmWave radar system and multi-protocol wireless connectivity, including BLE5.3/802.15.4/Thread/Matter. The chip is divided into four subsystems:

- mmWave subsystem: This module encompasses all high-performance radar RF/analog and baseband digital signal circuits. It also incorporates a linear frequency modulation pulse generator and a Tx-to-Rx state machine;
- Wireless RF subsystem: This module encompasses a comprehensive suite of multi-protocol wireless RF, modulation/demodulation, and baseband protocols. Data interacts with the application CPU via an internal data bus;
- Processing and application subsystem: includes a 32-bit RISC CPU for different application processing and task management;
- Power and clock subsystem: RS6xxx is a complex single-chip SoC with multiple functional systems. We utilize the on-chip integrated PMU system and clock subsystem for centralized management.

2.2 Application Block Diagram

The typical applications of RS6xxx millimeter wave mainly consist of Flash, DCDC, and 32M crystal oscillators, with the main contents as follows:

- The external passive crystal oscillator with a frequency of 32MHz provides a precise reference clock for the sensor chip;
- Provide 3.3V power supply to module VCC and VDDIO;
- V13 outputs 1.5V or accepts an external DCDC input of 1.3V;
- Communicate with the host computer through the UART interface, and download and program firmware to an external Flash;
- GPIO interface;
- 2.4G ANT antenna (optional).

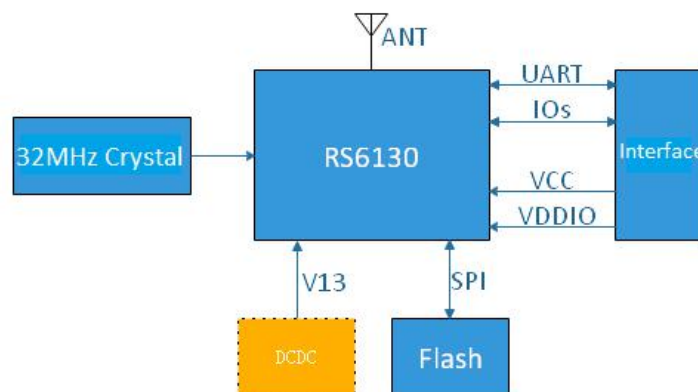


Figure 2-1 Application block diagram of RS6xxx

2.3 Production Testing System Development Guide

2.3.1 Composition of production testing system

To build a radar testing system, users need to develop the following four major types of components.

Table 2-1 Composition of Production Testing System

Control Code	The software engineer needs to write production test control code on the host computer to control the radar to perform relevant test actions. If the radar is an independent module, a PC-side tool will be used as the host computer to communicate with the radar through USB to UART/I2C/SPI conversion. Therefore, a conversion board needs to be designed, and correspondingly, the production test control code is implemented on the PC side. If the radar and the host MCU are connected and cannot be separated, the PC-side tool will communicate with the host MCU via UART to control the radar. Correspondingly, it is recommended to implement the production test control code on the MCU side.
Tooling Fixture	For items such as corner reflectors, corner reflector brackets, and radar brackets, due to the sensitivity of radar to specific materials and structures, tool design must adhere to the constraints provided by our company.
Control Hardware	Instrument: uA-level ammeter, used for power consumption testing and power supply. Communication connection: adapter board, such as USB to UART, I2C, SPI.
Test Environment	Radar is sensitive to moving objects, vibrating objects, and large-area metal objects, so when selecting a production and testing site, relevant constraints must be adhered to.

2.3.2 Production Testing System Development Steps

Step 1: Develop the host computer control code

- All control commands are developed based on the HIF protocol defined by our company. Users can refer to [the instructions in Section 3.3.1 HIF](#).
- Before starting to write code, please read [Sections 3.1 Production Testing Process](#) and [3.2 Test Item Description](#) to understand the executable test cases. Users can delete test items according to actual needs.
- To facilitate users in writing control code, our company has provided a Python reference code based on UART communication. For usage instructions, please refer to [Section 3.3.2.4 Factory_Test_Script](#). The code can be found at the following path: [Data Package\Software Kit\03_Tool\factory_test_script.py](#).
- Due to the need to modify certain test case input parameters based on the actual user-configured settings, such as starting frequency and trigger distance, the command requires recalculation of the verification code after parameter modification. Therefore, users must repackage the command according to the HIF protocol. For this action, our company provides a "command generation tool" to assist customers in generating commands. For specific usage instructions, please refer to [Section 3.3.2 "Command Generation Tool"](#).

The tool path is:

[DataPackage\Software_Kit\03_Tool\ProductionTestAnalysisTool\MRS6xxx_ProductTestingAnalysisTool.zip](#).

Step 2: Develop the tooling fixtures and control hardware required for testing

- For details, please refer to [Section 3.4.2](#).
- The radar bracket and angle reflector bracket are required items, while other tooling fixtures need to be designed based on the test items selected in Step 1.
- To ensure the stability and reliability of the testing environment, it is recommended to rigidly connect the radar bracket with the angle reflector bracket, for example, using aluminum profiles, to ensure that the relative positional relationship between the radar and the angle reflector does not easily change.
- Design a communication adapter board between the host computer and the radar. If there is a need for power consumption testing, a microampere-level precision ammeter should be prepared.

Step 3: Determine the testing environment

- Radar is sensitive to moving objects, vibrating objects, and large-area metal objects, so when selecting a production and testing site, relevant constraints must be observed.
- For specific requirements of environmental constraints, please refer to [Section 3.4.2](#).

Step 4: Acceptance testing environment

- Utilize the development board provided by our company, along with the performance test data based on the development board, to confirm whether the selected environment meets the requirements.
- For specific acceptance methods, please refer to [Section 3.4.3](#).

Step 5: Establish testing standards

- Refer to [Section 3.2.7 Test Item Indicators](#) to establish the test pass criteria.
- The formulation of radar performance indicators is relatively special, requiring users to combine factors such as their own hardware, structural components, environment, and set radar parameters. Only after collecting data from multiple prototypes and summarizing it can the pass criteria be formulated. For specific details, refer to [Section 3.3.1.9, "Method for Determining Performance Test Indicators"](#).

3 Radar System Testing

3.1 Production Testing Process

After the main controller or the host computer turns on the radar power switch, it waits for the radar to complete power-up. Then, it sends a production test mode command to activate the production test function, conducts production test items, and sends a production test mode command to disable the production test function after the test is completed. The operation process of production testing is as follows:

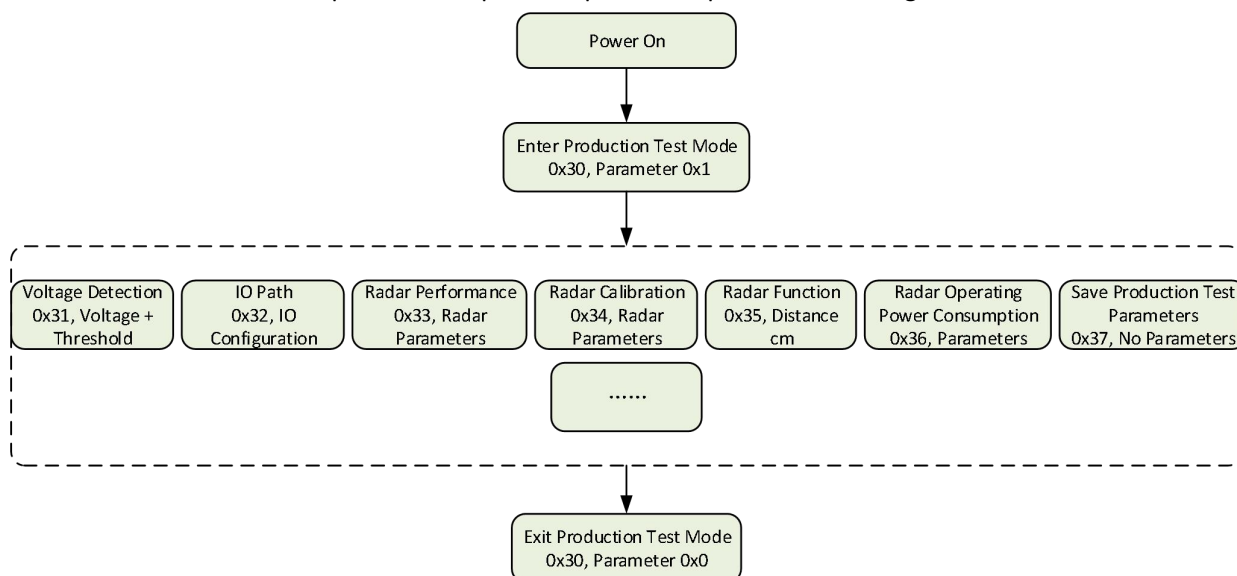


Figure 3-1 Radar starts the testing process

3.2 Test Item Description

This section primarily explains the meanings of various test items, while the next section will provide a detailed introduction on how to develop these test items.

3.2.1 Voltage measurement

The radar requires V13 to be powered by either an external DCDC or an internal LDO. If an external DCDC is used, the V13 voltage can be collected through the radar's internal ADC to check whether the external DCDC is functioning properly. If the radar is powered by an internal LDO, this test can be skipped as the chip has already been tested at the factory.

For scenarios where the V13 voltage cannot be tested through a probe after the entire machine is assembled, this method can also be used for testing.

3.2.2 IO path test

In practical radar applications, there are usually multiple signals connected to the host terminal. During production, this test can be used to check whether the signal functions are normal.

There are three types of test modes: IO docking, IO output, and IO input.

3.2.3 Radar performance testing

Radar performance testing refers to the testing of radar antenna transmission and reception performance indicators, including the average noise floor value, maximum noise floor value, minimum noise floor value, average level value (angle return power), and average SNR (signal-to-noise ratio) of each receiving antenna.

Radar performance testing has strict requirements for the testing environment. When selecting a testing site and setting up a testing platform, relevant regulations must be adhered to. For specific details, please refer to the description of "Testing Environment Requirements" in the following text.

The performance of radar is significantly affected by its housing. Therefore, if the product has a housing, it needs to be tested with the housing assembled during performance testing.

3.2.4 Radar angle correction

Radar has the function of angle measurement, and its angle measurement error is affected by antenna performance and external structural components. Therefore, during the production of actual complete products, angle correction operations need to be performed after assembling structural components (especially those that are obstructing the radar surface).

After the angle correction is passed, the calibration values need to be saved to the flash memory on the radar side.

3.2.5 Radar function test

The radar chip enters the trigger test mode, and upon detecting a moving object, PA0 outputs a high level. The high level of PA0 is maintained for 1 second before outputting a low level.

This test item allows for verification of whether the radar system is functioning properly from the application level.

3.2.6 Power consumption test in radar working scenario

Under normal operating conditions, the radar may exhibit different operating states due to the influence of external moving targets, resulting in varying power consumption. Therefore, to detect whether the power consumption of the machine is normal during production, this test item can be utilized. After executing this test, the radar will be stabilized in a specific operating state, protected from external influences.

3.2.7 Test item indicators

The following are the indicator standards that need to be controlled during testing.

Table 3-1 Test Item Indicator Standards

Type	Project		Return Value Type	Specifications			Unit
				Min	Typ	Max	
Functional Testing	Voltage measurement (mandatory for devices with DCDC function)		Numerical Value	A-75	A ¹	A+75	mV
	IO pathway ⁵		Status/Value	PASS/0, 1			-
	Radar performance (three groups)	Average noise floor	Numerical Value	B-600	B ²	B+600	-
		Minimum value of noise floor at a single point (without card control ⁶)	Numerical Value	C-4000	C ²	C+4000	-
		Maximum value of single point of background noise	Numerical Value	D-1000	D ²	D+1000	-
		Level value	Numerical Value	E-500	E ²	E+500	-
		SNR value	Numerical Value	F-500	F ²	F+500	-

	Radar calibration	Calibration	Numerical Value	G-2	G ³	G+2	°
	Radar function	-	State	IO high level indication			
	Power consumption in radar operating scenarios		Numerical Value	H-0.2	H ⁴	H+0.2	mW

- 1) According to the customer's scheme, the value of A is determined based on the voltage. When using an external DCDC, it is 1.3V;
- 2) For radar performance values, it is necessary to collect test data from 10 machines in an actual testing environment. After confirming that the data are basically consistent, the average values of B, C, D, E, and F can be obtained. For specific collection methods, please refer to Section 3.3.1.9 "Method for Determining Performance Test Indicators";
- 3) The default radar angle correction is 0°, which corresponds to G = 0;
- 4) Collect the power consumption data of 10 machines in their working scenarios and calculate the average to obtain H. For specific collection methods, please refer to Section 3.3.3.13 "Method for Determining Scene Power Consumption Indicators";
- 5) When selecting IO docking mode or IO input mode for testing, the output result is PASS/FAIL. When selecting IO output mode for testing, the IO output voltage is collected by the main controller or the host computer.
- 6) The minimum value of the noise floor at a single point is not controlled during actual testing, but it still needs to be configured when setting the parameters.

3.3 Test Case Software Development

This section introduces how to construct test cases and provides guidance for software developers in developing test code.

There are two platforms for the development of this radar software, namely the SDK development platform and the no-development platform.

The SDK development platform is based on APIs for development, while the no-code development platform is based on HIF (Host Interface) for development. Therefore, the following introduction to the software development method for test cases is also divided into two sections. Users are advised to select the corresponding section based on the platform they are using.

3.3.1 HIF

3.3.1.1 Overview

HIF, an abbreviation for Host Interface, is a set of customized communication protocols. For specific definitions, please refer to other documents in the resource package, such as the "RSxxx_xxxxHOST_Development Manual".

Production testing is a type of testing conducted in production testing mode, which is different from the normal working mode:

1. The production test mode requires special commands to enter and exit;
2. In production testing mode, only production testing-related commands can be sent;
3. After exiting the production test command, enter the normal working mode.

During production testing, it is necessary to wait for the system to finish starting up. Once the system has finished starting up, it will automatically report the Init_State_Report, as follows:

A5 48 12 FF 01 00 00 ED 00 FE FF

FF reports the MsgId for the initialization status, with a status code of 00 indicating successful system loading.

As shown in the figure below, the typical power-on time is approximately **2.5s to 3s**.

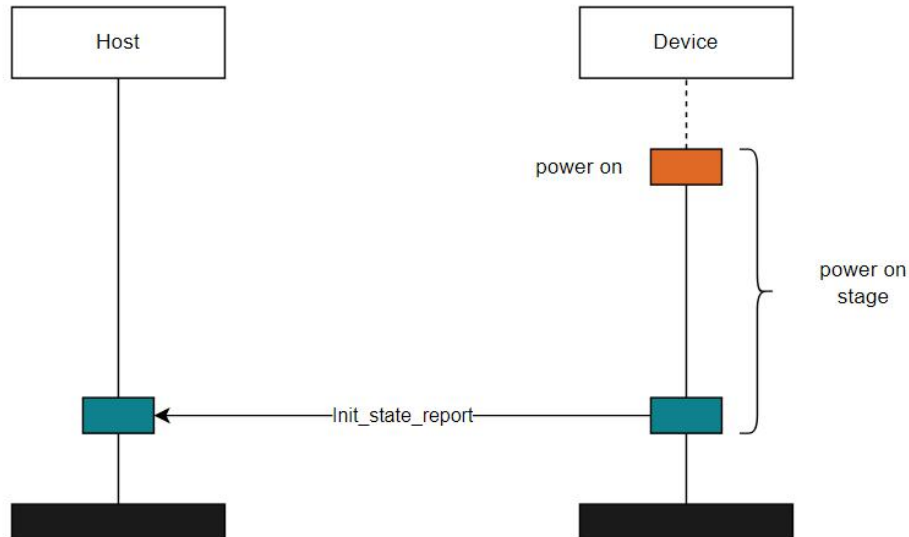


Figure 3-2 Power-on Process

Only after the system has been fully booted can you issue a command to enter the production test mode. The process is as follows, and the following items should be noted:

1. Before sending the "Enter_Factory_Test" command, it is necessary to wake up the device. The method is to send 0x55 0xFF repeatedly until 0x79 is returned. It is recommended that the loop interval be between 10ms and 100ms.
2. After waking up the device, the "Enter_Factory_Test" command needs to be sent within the "wake timeout" period; otherwise, the device will go into hibernation again and cannot correctly accept commands. The default wake timeout value is 2 seconds.
3. During the production testing process, status updates or data returns are only provided after each command has been executed. The execution time for each command varies, and it is recommended to set a maximum timeout of 10 seconds.
4. During the production testing process, each command corresponds to a reply. Sending the next command before a reply is received or before the timeout period is exceeded is not supported.

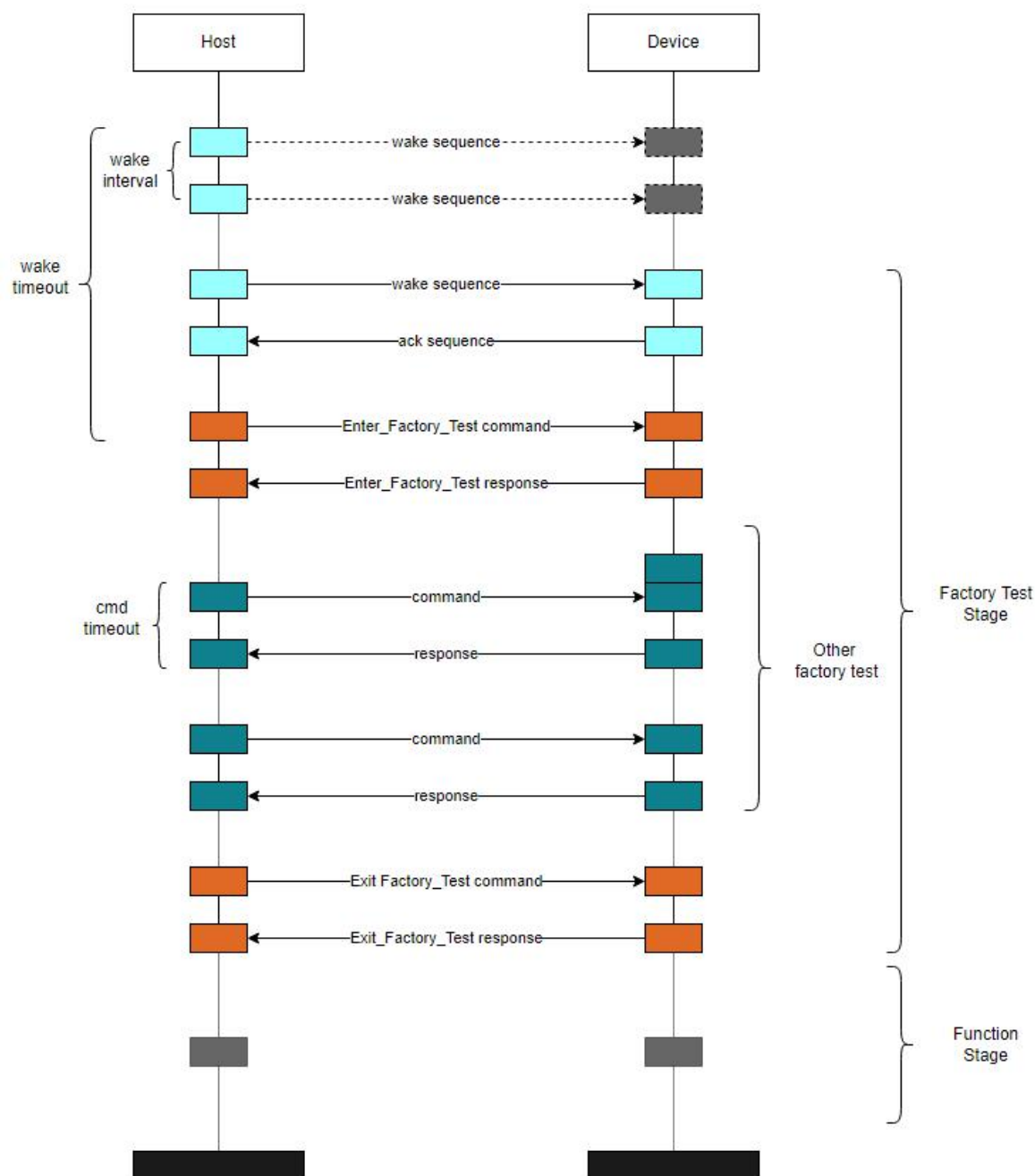


Figure 3-3 Schematic diagram of static configuration generation

3.3.1.2 Command example

The following text will explain the composition and usage precautions of each command. Commands are divided into two types:

- Type 1: Fixed commands that require no parameter modification and can be directly used by users, such as entering production testing/obtaining ID, etc.
- Type 2: Commands with parameters, where users need to modify the parameters according to instructions to generate new commands, such as performance testing/IO path testing.

To facilitate users in generating new commands and writing test cases, we have provided a command generation tool as well as sample code. For details, please refer to the "Command Generation Tool" section of this section.

Command generation tool: ProductTestingAnalysisTool, stored in SDK/Software_Kit/03_Tool

Example code: `factory_test_script.py`, stored in `SDK/Software_Kit/03_Tool`

1. Enter production test mode

CMD: A5 14 15 30 01 00 01 E9 CF FE FF

RSP: A5 07 12 30 01 10 00 ED CF FE EF

Judgment criteria: A complete match of ACK indicates success (a seventh bit of 00 indicates pass, otherwise it indicates false)

2. Voltage test // Optional, only for DCDC scheme

CMD: A5 AC 15 31 08 60 C9 04 00 00 5F 05 00 00 C2 C4 F7 9F

RSP: A5 06 12 31 01 10 00 ED CE FE EF

Judgment criteria: A complete ACK match indicates a pass (a seventh bit of 00 indicates a pass, otherwise it indicates false)

3. Performance testing // Data from 10 devices needs to be collected for parameter updates in the early stage

CMD:A5 8E 15 33 44 40 07 00 4C 00 E6 00 58 1B 50 00 54 EC 00 00 00 00 00 00 00 00 00 00 00 00 9A
E4 D2 E0 02 E4 4C FD 0E 19 73 E4 A6 E0 F5 E3 EB FC C4 18 25 E5 A6 E0 31 E4 78 FC 0D 18 F4 01 04 00 F4 01
90 01 90 01 E8 03 CD 29 A4 11

RSP: A5 8E 12 33 27 60 00 07 00 00 10 DA 04 34 E9 BE E4 A6 E0 17 E4 48 FD FB 18 9A E4 40 E0 06 E4 F0 FC
AE 18 CE E4 A6 E0 41 E4 85 FC 12 18 5B C4 33 FF

Judgment criteria: First, determine whether the ACK in the response starts with A5. Then, if the seventh bit is 00, it indicates pass; otherwise, it indicates false

4. Radar calibration

CMD: A5 C7 15 34 0A 40 07 00 58 1B 00 00 00 00 00 00 E3 CB 9D A4

RSP: A5 77 12 34 2D 70 00 00 00 DA 04 00 00 00 00 00 00 00 00 00 00 00 A5 07 00 00 C8 04 00 00 D9 00
98 06 03 03 00 08 EE 06 60 06 00 08 00 00 00 10 C9 3C 99 94 A3 F4

Judgment criteria: First, determine whether the ACK response starts with A5. Then, if the seventh bit is 00, it indicates a pass; otherwise, it indicates a false

5. Calibration value programming

CMD: A5 AE 15 37 00 60 EA C8 FF 9F

RSP: A5 E0 12 37 01 30 00 ED C8 FE CF

Judgment criteria: A complete match of ACK indicates success (if the seventh bit is 00, it indicates pass; otherwise, it indicates false)

6. IO path testing

CMD: A5 10 15 32 03 00 00 02 03 EA CB F9 FF**RSP:** A5 D5 12 32 01 40 00 ED CD FE BF**Judgment criteria:** A complete match of ACK indicates success (with the seventh bit being 00 indicating pass, otherwise indicating false)

7. Power consumption test in working scenarios

CMD: A5 0D 15 36 02 00 01 01 E9 C8 FD FF**RSP:** A5 0D 15 36 02 00 01 00 EA C8 FD FF**Judgment criteria:** A complete match of ACK indicates success (if the seventh bit is 00, it indicates pass; otherwise, it indicates false)

8. Exit production testing mode

CMD: A5 F4 15 30 01 20 00 EA CF FE DF**RSP:** A5 F7 12 30 01 20 00 ED CF FE DF**Judgment criteria:** A complete match of ACK indicates success (if the seventh bit is 00, it indicates pass; otherwise, it indicates false)**3.3.1.3** Enter production test mode

Note:

1. Only after this instruction can other production testing instructions be sent;
2. The device must be woken up before sending this command;
3. Before sending the "Exit Production Test Mode" command, if this command is sent repeatedly, it will be ignored (with a reply status of 0x0C);
4. After the device is restarted, it will exit the production test mode and enter the normal working mode;

CMDA5 14 15 **30** 01 00 **01** E9 CF FE FF**30** represents the entry or exit of the production test mode using MsgID, and the **01** parameter indicates entering the production test mode.**RSP**A5 07 12 **30** 01 10 **00** ED CF FE EF**30** is the MsgID for entering or exiting the production test mode. A status code of **00** indicates successful entry into the production test mode, while a status code of **0C** indicates that the device is already in production test mode and will ignore this command.**3.3.1.4** Exit production test mode

Note:

1. This command is sent after the production test is completed. After executing this command, the device enters normal function mode and cannot send production test commands anymore;
2. Repeatedly sending this command will be ignored (with a reply status of 0x0C);

CMD

```
A5 F4 15 30 01 20 00 EA CF FE DF
```

30 is the MsgID for entering or exiting the production test mode, and the 00 parameter indicates exiting the production test mode.

RSP

```
A5 F7 12 30 01 20 0C ED CF FE DF
```

30 is the MsgID for entering or exiting the production test mode. A status code of 0C indicates successful exit from the production test mode, while a status code of 00 indicates that the device is already in normal working mode.

3.3.1.5 Get ID

Obtain the CHIP ID of the system, with an ID size of 16Bytes. If the host computer needs to record the test data for each machine, the CHIP ID can be used as an identification code.

CMD

```
A5 43 15 01 01 00 01 E9 FE FE FF
```

01 is the MsgID for obtaining system information, and parameter 01 indicates obtaining the CHIP ID.

RSP

```
A5 26 12 01 11 10 00 3D B0 5C 0C 42 01 00 06 00 00 20 01 00 00 00 00 DA 7F 3D 73
```

01 is the MsgID for obtaining system information. The status code 00 indicates successful retrieval, and the return value is 3D B0 5C 0C 42 01 00 06 00 00 20 01 00 00 00 00 DA, representing the CHIP ID.

3.3.1.6 Voltage measurement

Voltage measurement is required for using the DCDC function in the module. The command inputs the desired minimum and maximum voltage values, and after testing, the current voltage value and status are returned.

The voltage unit is mV. It is recommended to refer to the minimum and maximum voltages of 1225mV and 1375mV respectively.

CMD

```
A5 AC 15 31 08 60 C9 04 00 00 5F 05 00 00 C2 C4 F7 9F
```

31 represents the voltage measurement MsgID, with parameter C9 04 00 00 indicating that the expected minimum voltage is not lower than 1225mv, and parameter 5F 05 00 00 indicating that the expected maximum voltage is not higher than 1375mv. (0x000004C9 = 1225, and 0x0000055F = 1375)

RSP

```
A5 F2 12 31 05 20 00 0A 05 00 00 ED C4 F5 DF
```

31 is the voltage measurement MsgID. The status code 00 indicates that the measured voltage is within the range, and the return value 0A 05 00 00 indicates that the current voltage is 1.29mV (0x0000050A = 1290).

RSP-

```
A5 D2 12 31 05 40 0C D7 05 00 00 E1 F7 F4 BF
```

31 is the MsgID for voltage measurement. The status code **0C** indicates that the test voltage is outside the target voltage range, and the return value **D7 05 00 00** indicates that the current voltage is 1.495mV (0x000005D7 = 1495).

If you need to change the voltage upper and lower limits, you can automatically generate instructions through a tool. For details, please refer to the "Command Generation Tool" in this section.

3.3.1.7 IO path test

The IO path tests the hardware connectivity, currently supporting three modes: IO docking, IO output, and IO input. The appropriate mode can be selected based on the actual situation.

1. IO docking test

Two IOs are short-circuited, with one serving as the output and the other as the input. The output IO outputs high and low levels respectively, while the input IO performs detection and judgment.

Parameters: 00 [input_IO] [output_IO] [input_pull]

input_IO/output_IO: 0~15 represent PA0~PA15, while 16~23 represent PB0~PB7.

input_pull: Represents the pull-up/pull-down state of input_IO. 0 indicates neither pull-up nor pull-down, 1 indicates pull-up, and 2 indicates pull-down.

CMD

```
A5 0F 15 32 04 00 00 05 07 02 EA C8 F4 FD
```

32 represents the IO path test MsgID. Parameter **00** indicates the IO docking test mode, parameter **05** indicates the input IO is PA5, parameter **07** indicates the output IO is PA7, and **02** indicates the input pin is pulled down.

RSP

```
A5 D5 12 32 01 40 00 ED CD FE BF
```

32 represents the IO path test MsgID. A status code of **00** indicates that the input IO and output IO are connected and functioning normally, while a status code of **0C** indicates that the input IO and output IO are connected but functioning abnormally.

2. IO output test

Specify the IO output as high or low level, **and use an instrument for testing and comparison.**

Parameter: 01 [output_IO] [output_value]

output_IO: 0~15 represent PA0~PA15, while 16~23 represent PB0~PB7.

output_value: 0 indicates low voltage level, while 1 indicates high voltage level.

CMD

```
A5 10 15 32 03 00 01 07 00 E9 C6 FC FF
```

32 represents the IO path test MsgID, with parameter **01** indicating the IO output test mode, parameter **07** indicating that the output IO is PA7, and parameter **00** indicating output at low level.

RSP

```
A5 C5 12 32 01 50 00 ED CD FE AF
```

32 is the IO path test MsgID. The status code **00** indicates successful command execution, but the voltage level on the IO needs to be measured by an instrument. If it is **0C**, it indicates command execution failure.


```

uint16_t e;           // Default value 230
uint16_t d;           // 0x1B58=7000, needs to be configured according to the selected
                        // maximum trigger distance, unit: millimeters
uint16_t dr;          // 0x0050=80, needs to be configured according to the selected
                        // distance resolution, unit: millimeter
uint32_t sf;          // 0xEC54=60500, needs to be configured according to the selected
                        // starting frequency, unit: MHz
uint32_t v;           // Default value 0
uint16_t vr;          // Default value 0
uint16_t f;           // Default value 0
uint32_t i;           // Default value 0
struct n_factor_t factor[N]; // Threshold mean value struct, for RS6130, N is 3, for RS6240, N is 8
struct n_gap_param_t gap; // Threshold structure
} __packed;

struct n_factor_t {
    // Radar performance Typ value (highlighted in red, requires
    // testing on 10 machines for averaging)
    int16_t D;          // Maximum mean value
    int16_t C;          // Minimum value average
    int16_t B;          // Average mean value
    int16_t E;          // Mean value of level 1
    int16_t F;          // Mean value of level 2
} __packed;

struct n_gap_param_t {
    // Radar performance threshold (F4 01 04 00 F4 01 A0 0F 58 02 E8 03)
    int16_t g0;          // Default value 500 (0x01F4), E±500
    int16_t g1; // default value 4 (0x0004)
    int16_t g2; // default value 500 (0x01F4), F±500
    int16_t g3; // default value 4000 (0x0FA0), C±4000
    int16_t g4;          // Default value 600 (0x0258), B±600
    int16_t g5; // default value 1000 (0x03E8), D ± 1000
} __packed;

```

RSP

```

A5 8E 12 33 27 60 00 07 00 00 10 DA 04 34 E9 BE E4 A6 E0 17 E4 48 FD FB 18 9A E4 40 E0 06 E4 F0 FC AE 18
CE E4 A6 E0 41 E4 85 FC 12 18 5B C4 33 FF

```

33 represents the radar performance test MsgID, and the status code 00 indicates a successful test. If the status code is 0C, it indicates an abnormal radar test. You can check whether the test environment is abnormal, whether the test instruction parameters are abnormal, and whether the chip operation is abnormal. If the status code is 0B, it represents an abnormal radar firmware system. You need to check the power supply and whether the img version is used incorrectly. Whether successful or failed, test data will be returned for analysis.

```

struct n_result {
    uint8_t m;
    uint8_t r_flag;
    uint8_t invalid_msk;
    uint8_t r;
    uint16_t d;
    int16_t n;
}

```

```
struct n_factor_t factor[3];    //Test results
} __packed;
```

3.3.1.9 Method for determining performance test indicators

Radar performance is strongly correlated with the design of the scheme's hardware and the testing environment. Therefore, the performance indicators for radar production testing need to be determined based on actual production testing data. First, select a trial production prototype for testing and data collection. Calculate and average the collected data to obtain the indicator values. The operation steps are as follows. The following is an example of HIF operation. Users who develop based on the SDK can configure similarly:

- 1) Send a performance data collection command to gather data. Please use the command generation tool to create a "data collection command" based on the radar configuration parameters. For specific details, refer to Section 3.3.2.2.4 "Generating Radar Performance Test Instructions".

- 2) Receive the data packet of radar performance test results.

Example of received data packet: A5 BE 12 33 27 30 0C 07 07 00 10 B0 09 8F E9 9B E5 BA E2 53 E4 27 F9 18 14 FF E5 8D E2 7F E4 A6 FA 44 15 A0 E5 3D E3 9E E4 1C FB D0 15 51 C8 2C 3F

Save the received data packet in its entirety.

Note: The red part represents the three sets of data in the factor two-dimensional structure of n-param.

- 3) Repeat steps 1) and 2) above, test 10 prototypes, and collect test data packets.
- 4) Analyze the test result data to obtain the radar performance production test indicators.

Sum up the D values in factor[0] from the test results of 10 prototypes and calculate the average to obtain the D index value of factor[0]. Similarly, calculate the C, B, E, and F index values. The calculation methods for D, C, B, E, and F in factor[1] and factor[2] are the same as in factor[0]. **This step can be done using a tool to interpret the contents of the data packet.**

- 5) Assess the rationality of the data from 10 prototype machines.

Compare the test results of the 10 prototypes with the indicator values calculated in step 4. The individual data must fall within the vicinity of the indicator values, with an interval range of $D \pm 500$, $C \pm 4000$, $B \pm 300$, $E \pm 250$, and $F \pm 250$. Otherwise, it indicates poor consistency of the prototypes, and the calculated results cannot be used as the indicator values for B, C, D, E, and F. New prototypes need to be selected to replace the boards with large deviations for retesting and recalculating the indicator values.

- 6) Generate performance test instructions.

Copy the test data package of 10 machines collected in Step 3 into the tool, and the performance test instructions can be automatically generated. For details, please refer to Section 3.3.2.2.4 "Generating Radar Performance Test Instructions".

Note:

- 1) Regarding the usage of the tool, please refer to the "Command Generation Tool" in this section.
- 2) The aforementioned 10 prototypes should be preserved as gold samples. When the testing environment changes, such as moving fixtures or changing the venue, the radar performance index values need to be redetermined.

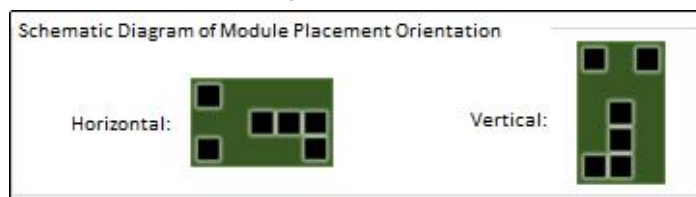
3.3.1.10 Radar calibration

The radar calibration test requires the establishment of a test environment according to the requirements. For specific details, refer to section 3.4.2 "Environment Setup".

Installation direction diagram of RS6130:



Installation direction diagram of RS6240:



CMD

A5 C7 15 **34** 0A 40 **07 00 58 1B 00 00 00 00 00 00** E3 CB 9D A4

34 represents the radar calibration MsgID, and the parameters **07 00 58 1B 00 00 00 00 00 00** are the radar calibration configuration, which is the content of the following struct.

```
struct a_param {
    uint8_t w;           // Configure 0x7 to perform calibration and angle measurement, configure
                        // 0x6 to perform angle measurement
    uint8_t t;           // Default configuration is 0, representing the installation direction of the
                        // radar, where 0 indicates horizontal and 1 indicates vertical
    uint16_t md;         // Default value 20480, unit: millimeters, configured according to the
                        // actual maximum trigger distance
    uint16_t dr;         // Default value: 80, unit: millimeter, configured according to actual
                        // distance resolution
    uint32_t sf;         // Default value 60500, unit: MHz, configured according to the actual
                        // starting frequency
} __packed;
```

RSP

A5 77 12 **34** 2D 70 **00** 00 00 DA 04 00 00 00 00 00 00 00 00 00 00 00 A5 07 00 00 C8 04 00 00 D9 00 98
06 03 03 00 08 EE 06 60 06 00 08 00 00 00 10 C9 3C 99 94 A3 F4

34 represents the radar calibration MsgID, and the status code **00** indicates calibration success. If the status code is **0C**, it signifies an abnormal radar calibration, requiring investigation into whether the testing environment is abnormal, whether the test command parameters are abnormal, and whether the chip operation is abnormal. If the status code is **0B**, it represents an abnormal radar firmware system, requiring investigation into the power supply and img version. Whether successful or failed, test data will be returned for analysis.

```
struct a_result {
    uint8_t r_flag; // Antenna error flag
    uint8_t m;      // Antenna involved in calibration evaluation
    uint16_t r;     // Distance r / 1000 unit: m
```

```

int n; // Noise
int s; // Signal-to-noise ratio
int r_c; //rx consistency
int adg; // Horizontal angle, adg / 1024, unit: °
int edg; // Pitch angle adg / 1024 unit: °
uint32_t calib[N]; // Calibration data, where N is 5 for RS6130 and 8 for RS6240
} __packed;

```

Please automatically generate instructions through a tool based on the actual radar installation direction, maximum trigger distance, distance resolution, and starting frequency point. For specific details, please refer to the "Command Generation Tool" in this section.

3.3.1.11 Radar function test

Test the basic functions of the radar. The radar detection range can be specified. Within the detection range, if the radar detects a target, it will report the target data, indicating successful testing. The recommended testing method is to wave within the detection range.

1. Enter radar function test

CMD

```
A5 0F 15 35 01 00 01 E9 CA FE FF
```

35 is the MsgID for radar function testing, and parameter 01 indicates entering the radar function testing mode.

RSP

```
A5 12 12 35 01 00 00 ED CA FE FF
```

35 is the MsgID for radar function testing. A status code of 00 indicates that the radar function test has been initiated and is functioning normally. If the status code is 0C, it indicates that the test has failed to initiate or has already been initiated.

2. Exit radar function test

CMD

```
A5 0F 15 35 01 00 00 EA CA FE FF
```

35 is the MsgID for radar function testing, and the parameter 00 indicates exiting the radar function testing mode. **Attention: The device needs to be awakened first (e.g. by sending 55 FF wake-up)**

RSP

```
A5 12 12 35 01 00 00 ED CA FE FF
```

35 is the MsgID for radar function testing. A status code of 00 indicates that the radar function test has exited normally, while a status code of 0C indicates that the exit was unsuccessful or has already occurred.

3. Reporting of detection data

```
A5 6A 12 C8 06 10 90 01 14 16 3E EB 1F 4B E4 D9
```

C8 represents the MsgID for reporting point cloud data, while 90 01 14 16 3E EB represents the location information data.

3.3.1.12 Scenario power consumption test

The scene power consumption test supports deep sleep low-power consumption testing and 2Hz typical standby scene power consumption testing.

Note:

After receiving the power consumption test entry command, the radar will automatically enter low power consumption mode, and the host does not need to send the sleep permission command (a5 cf 15 05 01 70 01 e9 fa fe 8f). If the sleep permission command is sent, it will actually cause the radar to exit sleep mode.

1. Deep sleep low power consumption test

The radar system is in a deep sleep state (radar is not working). It supports two wake-up methods: sleep timeout wake-up and IO wake-up.

Parameters: 00 [wake_type] [wake_param].

"wake_type = 0" indicates IO wake-up, and "wake_param" specifies the IO to be woken up;

"wake_type = 1" indicates timeout wake-up, and "wake_param" specifies the wake-up timeout in seconds;

IO wakeup

CMD

```
A5 0C 15 36 03 00 00 00 0A EA C9 F2 FF
```

36 is the MsgID for power consumption testing. Parameter 00 indicates the deep sleep low-power consumption testing mode, parameter 00 indicates the IO wake-up mode, and parameter 0A indicates that the wake-up IO is PA10.

RSP

```
A5 F1 12 36 01 20 00 ED C9 FE FF
```

36 is the MsgID for power consumption testing. A status code of 00 indicates successful entry into low power consumption mode, and the power consumption value needs to be measured using an instrument.

Timeout wakeup

CMD

```
A5 0C 15 36 03 00 00 01 0A EA C8 F2 FF
```

36 is the MsgID for power consumption testing. Parameter 00 indicates the deep sleep low-power testing mode, parameter 01 indicates the wake-up mode as automatic wake-up upon timeout, and parameter 0A indicates a wake-up timeout of 10 seconds.

RSP

```
A5 F1 12 36 01 20 00 ED C9 FE FF
```

36 is the MsgID for power consumption testing. A status code of 00 indicates successful entry into low power consumption mode, and the power consumption value needs to be measured using an instrument.

2. Power consumption test in typical standby scenario at 2Hz

The radar operates at a frequency of 2Hz and requires a specific command to enter and exit this state.

Note:

Before sending the exit command, you need to wake up the device first (for example, send 0x55 0xFF until a reply of 0x79 is received).

Enter 2Hz power consumption test

CMD

```
A5 0D 15 36 02 00 01 01 E9 C8 FD FF
```

36 is the MsgID for power consumption testing. Parameter 01 indicates the power consumption testing method for a typical standby scenario at 2Hz, and parameter 01 indicates entry into this scenario.

RSP

```
A5 F1 12 36 01 20 00 ED C9 FE FF
```

36 is the MsgID for power consumption testing. A status code of 00 indicates successful entry into low power consumption mode, and the power consumption value needs to be measured using an instrument.

Exit 2Hz power consumption test

CMD

```
A5 0D 15 36 02 00 01 00 E9 C9 FD FF
```

36 is the MsgID for power consumption testing. Parameter 01 indicates the power consumption testing method for a typical standby scenario at 2Hz, while parameter 00 indicates exiting this scenario.

Attention: Before sending the exit command, the device needs to be awakened first (e.g. by sending 55 FF wake-up)

RSP

```
A5 F1 12 36 01 20 00 ED C9 FE FF
```

36 is the MsgID for power consumption testing, and the status code 00 indicates successful exit from this power consumption testing scenario.

If command parameters need to be modified, users can automatically generate instructions through tool configuration. For details, please refer to the "Command Generation Tool" in this section.

3.3.1.13 Method for determining scene power consumption indicators

Radar power consumption is strongly correlated with the hardware of the solution and the actual working scenario. Therefore, the power consumption indicators for radar production testing need to be determined based on actual test data. First, 10 prototypes are selected for testing and data collection. The collected data is averaged to obtain the indicator value. The operation steps are as follows:

- 1) The radar enters the power consumption test mode, and a high-precision measuring instrument (with current measurement accuracy of $\leq 1\mu A$) is used to measure the power consumption of VCC and VDDIO, and record the power consumption values.
- 2) The power consumption values of 10 prototype devices (with good consistency in power consumption and a deviation of $\leq 0.1mW$) are collected and averaged to obtain the power consumption index value H for the radar operating scenario.

3.3.1.14 Calibration programming

1. After successful radar calibration, the calibration data is stored in the SPI-Flash module through this command.

CMD

```
A5 0E 15 37 00 00 EA C8 FF FF
```

37 represents HIF msgID, which is the HIF command for radar calibration.

RSP

```
A5 00 12 37 01 10 00 ED C8 FE EF
```

37 indicates the calibration programming of MsgID. The status code 00 indicates successful data storage. If the status code is 0C, it indicates data storage failure, usually due to file system anomalies in the radar. It is necessary to check whether the SPI FLASH chip is on the supported list and whether the radar img is used correctly.

2.Import calibration data from the external source and store it in the SPI-Flash module.

CMD

```
A5 F9 15 37 15 00 01 87 06 03 FD 00 08 5D FD 4A 06 07 FC 00 08 00 00 00 10 09 3C B7 F4 BD CF
```

37 represents the HIF msgID, which is the HIF command for radar calibration. 01 represents the subid, and 87 06 03 FD 00 08 5D FD 4A 06 07 FC 00 08 00 00 00 10 09 3C represents the 20-byte calibration data.

RSP

```
A5 00 12 37 01 10 00 ED C8 FE EF
```

37 indicates calibration programming of MsgID. The status code 00 indicates successful data storage. If the status code is 0C, it indicates data storage failure, usually due to an abnormal file system in the radar. It is necessary to check whether the SPI FLASH chip is on the supported list and whether the radar img is used correctly.

3.3.1.15 Calibration data acquisition

This command is used to retrieve the radar calibration parameters stored in the SPI-Flash.

CMD

```
A5 0D 15 38 00 00 EA C7 FF FF
```

38 indicates the acquisition of radar calibration parameter MsgID.

RSP

```
A5 FB 12 38 15 00 00 F5 07 1B 00 00 08 12 FE A6 07 F3 FA 00 08 00 00 00 10 C9 3C B9 2A BB 16
```

38 indicates the acquisition of radar calibration parameter MsgID. Status code 00 indicates successful data acquisition, but the validity of the data requires analysis. If the status code is 0C, it indicates that calibration has not been performed.

3.3.2 Command generation tool

To facilitate users in generating new commands and writing test cases, we have provided a command generation tool as well as sample code.

Command generation tool: ProductTestingAnalysisTool, stored in SDK/Software_Kit/03_Tool, used to interpret HIF data packets and generate commands.

Example code: factory_test_script.py, stored in SDK/Software_Kit/03_Tool. Users can generate commands based on this code using the "Command Generation Tool", replacing the commands in the example code to form their own production testing code.

3.3.2.1 ProductionTestAnalysisTool

The ProductionTestAnalysisTool is used to parse and convert HIF production test instructions and data. It also supports generating HIF instructions required for production testing based on different parameter configurations, and enables the measurement of ambient noise through this tool.

3.3.2.1.1. Transition zone

3.3.2.1.1.1. Analyze production test performance data

- (1) Input the HIF protocol data for performance testing (e.g., A5 AE 12 33...), and paste it into the HIF message input text box, ensuring it is complete and accurate;
- (2) Press the conversion button, and the parsed data will be displayed in the HIF message conversion result text box, as shown below;

The screenshot displays the MRS6130 software interface, which is used for radar system testing. The interface is divided into several sections:

- 解析 (Parse):** Contains a text box for HIF message input and a "转换" (Convert) button. Below it, the "HIF消息转换结果" (HIF Message Conversion Result) is displayed, showing a successful result with various performance metrics for Antenna1 and Antenna2.
- 生成区 (Generation Area):** Contains multiple sections for generating test commands:
 - 电压测量 (Voltage Measurement):** Includes input fields for minimum and maximum voltage values and a "生成电压测量指令" (Generate Voltage Measurement Command) button.
 - IO通路测试 (IO Path Test):** Includes a dropdown for test mode, input/output selection, and a "生成IO通路测试指令" (Generate IO Path Test Command) button.
 - 场景功耗测试 (Scenario Power Consumption Test):** Includes a dropdown for test mode, wake-up mode, and a "生成功耗测试指令" (Generate Power Consumption Test Command) button.
 - 雷达性能测试 (Radar Performance Test):** Includes sliders for start frequency, distance resolution, and trigger distance, along with a "生成雷达性能测试指令" (Generate Radar Performance Test Command) button.
 - 雷达校准 (Radar Calibration):** Includes a dropdown for radar installation direction, a dropdown for work mode, and a "生成角度测试指令" (Generate Angle Test Command) button.

Figure 3-4 Example diagram of performance data

3.3.2.1.1.2. Analyze production test angle data

- (1) Input the HIF protocol data for angle testing (e.g., A5 A7 12 34 ...), and paste it into the data conversion box, ensuring it is complete and accurate;
- (2) Press the conversion button, and the parsed distance and angle data of the antenna will be displayed in the conversion result box.

解析 **执行**

转换区

HIF消息输入

```
A5 A7 12 34 2D 40 00 00 00 B0 04 00 00 00 00 00 00 00 00 00 00 00 00 2B
FD FF FF A0 FD FF FF D7 06 3D 00 00 08 00 08 0F 07 2A FF 00 08 00 00 00
10 C9 3C B0 97 A9 DF
```

● 解析 ○ 角度数据采集 **转换**

HIF消息转换结果

```
Result: SUCCESS!
Range: 1.2m
Azimuth_degree: -0.7080078125
Elevation_degree: -0.59375
Antenna0_calib_I: 0x6D7 1751
Antenna0_calib_Q: 0x3D 61
Antenna1_calib_I: 0x800 2048
Antenna1_calib_Q: 0x0 0
Antenna2_calib_I: 0x78F 1935
Antenna2_calib_Q: 0xFF2A -214
Antenna3_calib_I: 0x800 2048
Antenna3_calib_Q: 0x0 0
Antenna4_calib_I: 0x1000 4096
Antenna4_calib_Q: 0x3C9 15561
```

生成区

电压测量

最小电压值(v) 1.225 最大电压值(v) 1.375

A5 DC 15 31 08 30 C9 04 00 00 5F 05 00 00 C2 C4 F7 CF **生成电压测量指令**

IO通路测试

测试方式 IO对接测试 输入IO PA0 输入IO上下拉 上拉 输出IO PA1

A5 CF 15 32 04 40 00 00 01 01 EA CD FA BE **生成IO通路测试指令**

场景功耗测试

测试方式 深度休眠功耗测试 唤醒方式 IO唤醒 唤醒IO PA0

A5 BC 15 36 03 50 00 00 00 EA C9 FC AF **生成功耗测试指令**

雷达性能测试

起始频点(GHz) 58 61 60.5 距离分辨率(cm) 5 12.5 8

触发距离(m) 0 20 7 实际最近探测距离(m): 10.24m

卡控阈值: 底噪平均值 600 底噪最小值 4000 底噪最大值 1000 电平值 500 SNR 500 **重置阈值**

性能数据输入框:

```
A5 9E 12 33 27 50 00 07 00 10 DA 04 62 E8 F9 E3 08 E0 34 E3 0E FC 98 18 7C E4 D2 E0 7E E3 AF FB F8 17 F4
E3 74 E0 38 E3 65 FB ED 17 60 86 38 67
A5 9E 12 33 27 50 00 07 00 10 DA 04 5D E8 EE E3 08 E0 38 E3 16 FC 9D 18 6A E4 A6 E0 68 E3 CD FB 1F 18 0D
E4 08 E0 39 E3 B1 FB 21 18 5F 48 38 85
A5 AE 12 33 27 40 00 07 00 10 DA 04 5E E8 08 E4 40 E0 33 E3 16 FC B3 18 91 E4 A6 E0 6C E3 CD FB 18 18 17
E4 40 E0 3F E3 B2 FB 28 18 5F 06 38 FF
```

E4 78 FC 0D 18 25 E5 A6 E0 31 E4 78 FC 0D 18 F4 01 04 00 F4 01 A0 0F 58 02 E8 03 CE 29 A6 37 **生成数据输入指令**

A5 BE 15 33 44 10 07 00 4C 00 E8 00 00 28 50 00 54 EC 00 00 00 00 00 00 00 00 00 00 **生成性能测试指令**

雷达校准

雷达安装方向 水平 工作模式 角度校准 最近探测距离(m) 20.48 起始频点(GHz) 58 距离分辨率(cm) 8

A5 E7 15 34 0A 20 07 00 00 50 50 00 90 E2 00 00 93 CB 65 AD **生成角度测试指令**

操作按钮: 清空 添加到性能数据采集 创建HIF消息

Figure 3-5 Example diagram of angle data

3.3.2.2 Generation Zone

3.3.2.2.1. Generate voltage measurement command

- (1) Modify the minimum DCDC voltage value (default 1.225v) and maximum DCDC voltage value (default 1.375v), and click to generate voltage measurement instructions.

电压测量

最小电压值(mv) 1.225 最大电压值(mv) 1.375

A5 EC 15 31 08 20 C9 04 00 00 5F 05 00 00 C2 C4 F7 DF **生成电压测量指令**

Figure 3-6 Example diagram of generating voltage measurement instructions

3.3.2.2.2. Generate IO path test instructions

The methods for selecting IO path testing include IO docking test, IO input test, and IO output test.

- (1) To test whether the IO connection is connected, select the test method as IO connection test, and choose the input and output PIN pins. Finally, click to generate the IO path test command (**note: the input pins need to be configured with pull-up, pull-down, or no pull-up/down as needed**);
- (2) To test whether the IO input is normal, select the test method as IO input test, and choose the pull-up/down method. Select the PIN pin and the input level value, and finally click to generate the IO path test instruction; (**note: the input pins need to be configured with pull-up, pull-down, or no pull-up/down as needed**);
- (3) To test whether the IO output is normal, select the test method as IO output test, choose the output PIN pin, select the output voltage level, and finally click to generate the IO path test command.

电压测量

最小电压值(v) 1.225 最大电压值(v) 1.375

A5 FC 15 31 08 10 C9 04 00 00 5F 05 00 00 C2 C4 F7 EF **生成电压测量指令**

Figure 3-7 Example diagram of generating IO path test instructions

3.3.2.2.3. Generate scenario power consumption test instructions

The methods for testing power consumption in scenarios include deep sleep power consumption testing and 2HZ typical scenario power consumption testing.

- (1) To measure deep sleep power consumption, select the test method as deep sleep power consumption test, and choose the wake-up method, which includes IO wake-up or timer wake-up. When selecting IO wake-up for radar, choose the wake-up PIN pin and click to generate the power consumption test command; when selecting timer wake-up for radar, you need to modify the timeout wake-up time and finally click to generate the power consumption test command.
- (2) To measure power consumption in a typical 2HZ scenario, select the test method as "2HZ typical scenario power consumption test" and directly click to generate the power consumption test instruction.

Figure 3-8 Example diagram of generating scenario power consumption test instructions

3.3.2.2.4. Generate radar performance test instructions

There are three configurations that need to be modified for testing radar performance commands. The furthest trigger distance, starting frequency point, and distance resolution should be adjusted according to specific applications.

- (1) Modify the maximum trigger distance, starting frequency point, and distance resolution according to actual requirements, and generate performance data collection instructions.
- (2) Use data collection instructions to collect performance test data from N units, which will be used for generating subsequent performance test instructions.
- (3) Paste the collected performance data into the performance data collection text box, separating each set of data with a newline character.
- (4) Adjust the card control threshold based on actual conditions. Optional
- (5) Click to generate performance test instructions and obtain the final performance test instructions.

Figure 3-9 Example diagram of generating radar performance test instructions

3.3.2.2.5. Generate angle calibration command

- (1) Configure the installation direction of the radar, selecting horizontal or vertical;
- (2) Calibration of angle selection in working mode;
- (3) Modify the maximum detection distance;
- (4) Modify the starting frequency point;

- (5) Modify the distance resolution according to the requirements
- (6) Click to generate angle calibration instructions.

雷达校准

雷达安装方向: 水平 | 工作模式: 角度校准 | 最远探测距离(m): 7 | 起始频点(GHz): 60.5 | 距离分辨率(cm): 8

A5 B7 15 34 0A 50 07 00 58 1B 50 00 54 EC 00 00 93 CB 49 A8

生成角度测试指令

Figure 3-10 Example diagram of generating angle test instructions

3.3.2.2.6. Generate angle measurement instruction

- (1) Configure the installation direction of the radar, selecting horizontal or vertical;
- (2) Select angle measurement for working mode;
- (3) Modify the maximum detection distance;
- (4) Modify the starting frequency point;
- (5) Modify the distance resolution according to the requirements
- (6) Click to generate angle measurement instructions.

雷达校准

雷达安装方向: 垂直 | 工作模式: 角度测量 | 最远探测距离(m): 20.48 | 起始频点(GHz): 58 | 距离分辨率(cm): 8

A5 D7 15 34 0A 30 06 01 00 50 50 00 90 E2 00 00 94 CA 65 9D

生成角度测试指令

Figure 3-11 Example diagram of generating angle measurement instructions

3.3.2.3 Measure the background noise level of the environment

- (1) Click the "Execute" button in the upper left corner of the tool to switch to the background noise measurement interface.
- (2) Connect to the development board, select the serial port number and baud rate, and click "connect".
- (3) According to the selected radar configuration, fill in the starting frequency point, distance resolution, and maximum trigger distance.
- (4) Click to enter "Production Test Mode", and then click "Noise Estimation" to obtain the noise measurement result.

Figure 3-14 Production Test Command Diagram

3.3.2.4.2. Execute script

3.3.2.4.2.1. Python environment

To ensure the proper functioning of FactoryTestScript, this machine needs to have a complete Python runtime environment and install the necessary third-party library pyserial to support serial communication.

3.3.2.4.2.2. Run the script

In the Python environment, open the command prompt (CMD) to execute the script: python factory_test_sample.py

3.3.2.4.3. View results

3.3.2.4.3.1. View in command line window

After executing the script through the command line, you can view the real-time test result data in the console window.

```
D:\Python_Test_SDK\Factory_Test>python factory_test_sample.py
2025-02-12 14:31:38,896 - INFO - ##### 角度校正 #####
2025-02-12 14:31:40,898 - INFO - Sent: 55ff
2025-02-12 14:31:40,998 - INFO - Sent: 55ff
2025-02-12 14:31:41,500 - INFO - Received: 79
2025-02-12 14:31:41,500 - INFO - 雷达唤醒成功
2025-02-12 14:31:41,523 - INFO - Sent: a5141530010001e9cffe
2025-02-12 14:31:41,925 - INFO - Received: A5071230011000EDCFFEEF
2025-02-12 14:31:41,926 - INFO - 进入产测模式成功!
2025-02-12 14:31:41,926 - INFO - Sent: a5c715340a400700581b000000000000e3cb9da4
2025-02-12 14:31:42,228 - INFO - Received:
2025-02-12 14:31:42,530 - INFO - Received:
```

Figure 3-15 Schematic diagram of script execution results

3.3.2.4.3.2. View log files

You can review the test logs through the logs, which will be synchronously saved to the same-level directory of the current script, named Factory_Test_log.

```
Factory_Test_20250212_14_31_38.txt - 记事本
文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)
2025-02-12 14:31:38,896 - INFO - ##### 角度校正 #####
2025-02-12 14:31:40,898 - INFO - Sent: 55ff
2025-02-12 14:31:40,998 - INFO - Sent: 55ff
2025-02-12 14:31:41,500 - INFO - Received: 79
2025-02-12 14:31:41,500 - INFO - 雷达唤醒成功
2025-02-12 14:31:41,523 - INFO - Sent: a5141530010001e9cffe
2025-02-12 14:31:41,925 - INFO - Received: A5071230011000EDCFFEEF
2025-02-12 14:31:41,926 - INFO - 进入产测模式成功!
2025-02-12 14:31:41,926 - INFO - Sent: a5c715340a400700581b000000000000e3cb9da4
2025-02-12 14:31:42,228 - INFO - Received:
2025-02-12 14:31:42,530 - INFO - Received:
```

Figure 3-16 Schematic diagram of Log

3.3.3 SDK

For customers developing based on SDK, an initialization interface is introduced in the factory test development reference case, which can be found at project/mmwave/factory_test. Developers only need to initialize the HIF driver and factory test HIF messages in their main function, and import the dependent files into the project to develop test cases. The files dependent on factory test development are located in the subsys/mmwave/mmwave_ft folder. If developers need to change the default parameters in the factory test process, they can modify the values of the corresponding macros in factort_test_config.h.

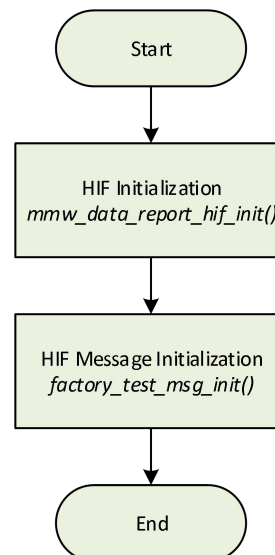


Figure 3-17 Flowchart of entering production testing by calling initialization API

3.3.3.1 System initialization API

3.3.3.1.1 mmw_data_report_hif_init

Table 3-2 mmw_data_report_hif_init API Brief

Classification	Instructions
Function	void mmw_data_report_hif_init(uint8_t mode, uint32_t rate, uint8_t type)
Function	Initialize the HIF communication mode and baud rate.
Parameter	uint8_t mode: Sets the communication mode to SPI or UART. uint32_t rate: Sets the baud rate of the selected communication method. uint32_t type: 1: need to transmit datacube; 0: no need to transmit datacube.
Return Value	Nothing
Note	If HIF has already been initialized, there is no need to call this HIF initialization interface. If this interface is called to initialize HIF, the type is generally set to 0.

3.3.3.1.2 factory_test_msg_init

Table 3-3 factory_test_msg_init API Brief

Classification	Instructions
Function	int factory_test_msg_init(void)
Function	Initialize the production test HIF message. In this function, multiple callbacks related to production test functions are registered. By distinguishing different msgids of HIF messages, different callbacks are entered to implement relevant functions.
Parameter	Nothing
Return value	Status value, 0 - Success, non-0 - Failure.

3.3.3.1.3 factory_test_callback_register

Table 3-4 factory_test_callback_register API Brief

Classification	Instructions
Function	void factory_test_callback_register(void (*usr_callback)(uint8_t event))
Function	Used for registering callbacks for entering/exiting production test mode. When entering/exiting production test mode, the usr_callback function pointed to by the function pointer (void (*usr_callback)(uint8_t event)) will be executed.
Parameter	void (*usr_callback)(uint8_t event) is a function pointer that points to the callback function to be executed. event: FT_EVT_EXIT, indicating exit from production test mode; FT_EVT_ENTER, indicating entry into production test mode;
Return Value	None.

3.3.3.2 Production testing function process

This section introduces the implementation process of the core module functions in the development of the production test function. If developers need to change the implementation process, they can modify the corresponding parts. For reference, please refer to /subsys/mmw/mmw_ft/factory_test.c.

3.3.3.2.1 Noise floor estimation

The implementation process of noise floor estimation is shown in the figure below, where the default number of transmitted waveform frames is 20.

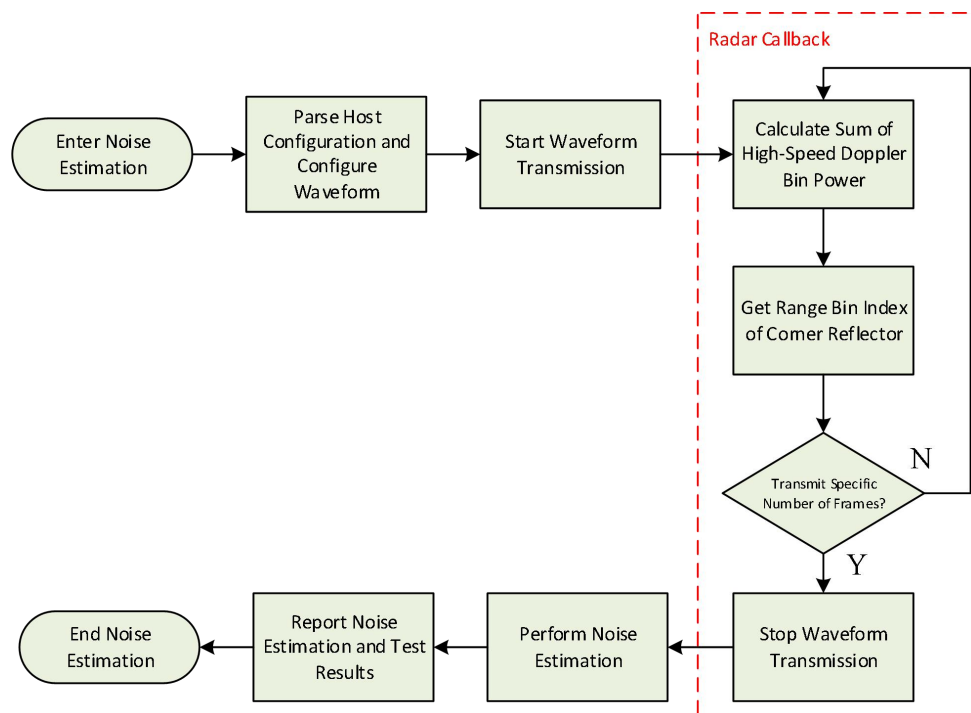


Figure 3-18 Flowchart of Noise Floor Estimation Implementation

The following figure illustrates the acquisition process of each frame of data on each receiving channel in the noise floor estimation process.

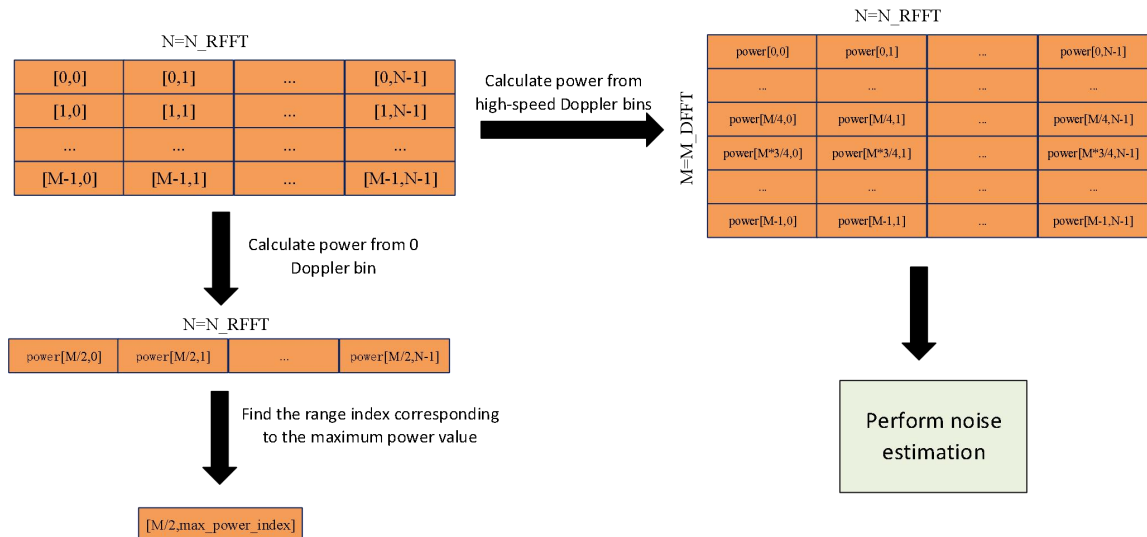


Figure 3-19 Schematic diagram of noise floor estimation data acquisition

3.3.3.2.2 Angle calibration and Angle measurement

The implementation process for angle calibration and angle measurement is shown in the figure below. The default number of transmitted waveform frames is 10. For **RS6130**, the calibration value in the antenna calibration register has already been multiplied when generating the Datacube. Therefore, in the angle calibration function, the antenna calibration register should be **disabled**, and after the angle calibration is successful, the angle calibration result should be written to the antenna calibration register. In the angle measurement function, the antenna calibration register should be **enabled**, and there is no need to perform antenna calibration before angle measurement. For **RS6240**, both the angle calibration function and the angle measurement function require **the antenna calibration register to be disabled**. Therefore, **antenna calibration is necessary before** angle measurement, which means that the IQ data should be multiplied by the calibration value before angle measurement can be performed.

It should be noted that:

1. After executing the angle calibration function, an angle measurement will be conducted immediately, followed by the return of both the angle measurement result and the calibration result.
2. If no angle calibration has been performed in the angle measurement function, the calibration values stored in flash will be used as the calibration values in the angle measurement function.

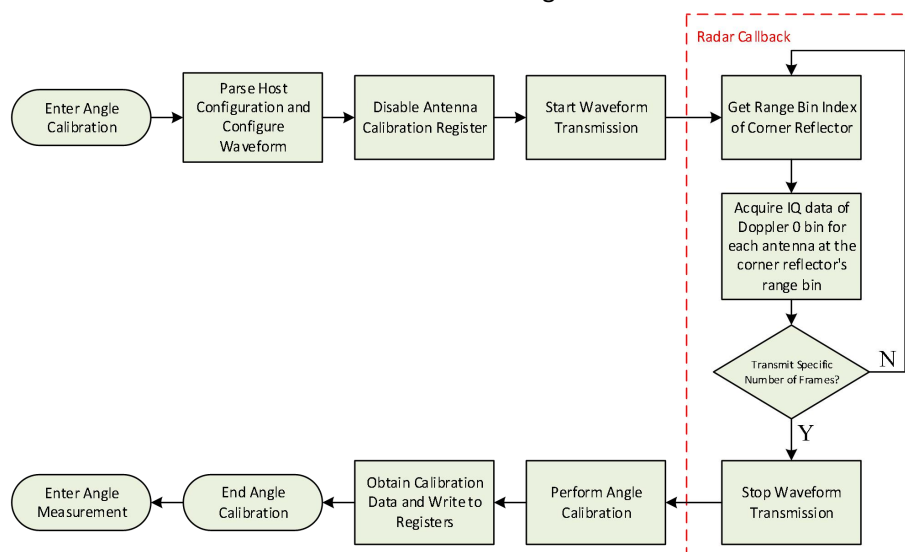


Figure 3-20 Flowchart of angle calibration implementation

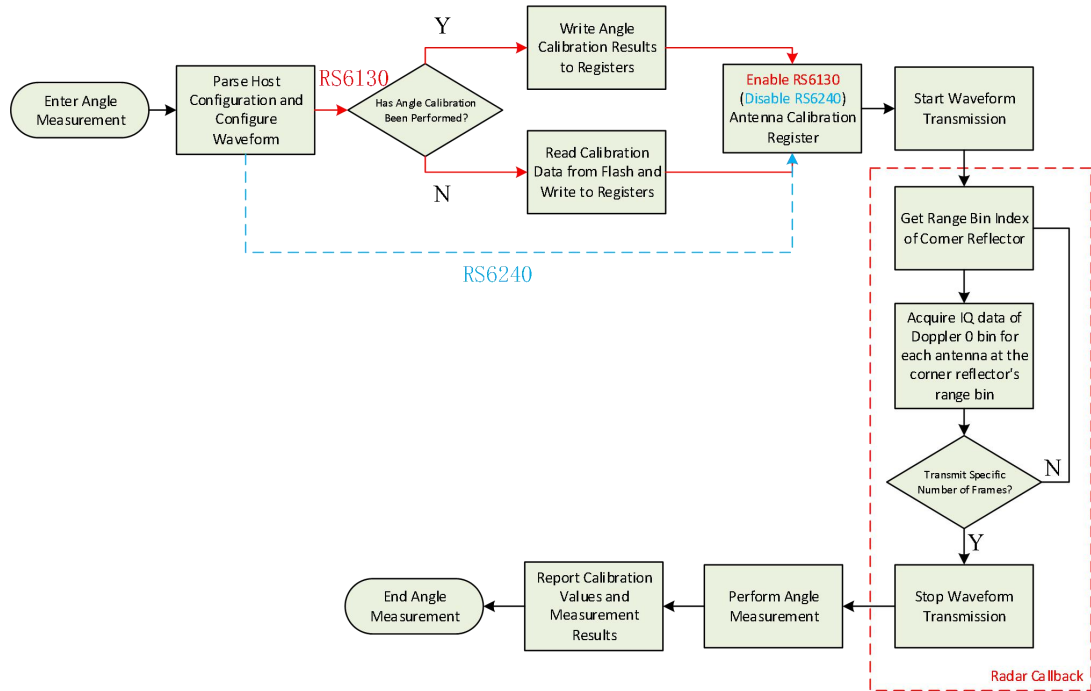


Figure 3-21 Flowchart of angle measurement implementation

The following figure illustrates the acquisition process of each frame of data on each receiving channel during the angle calibration/measurement process.

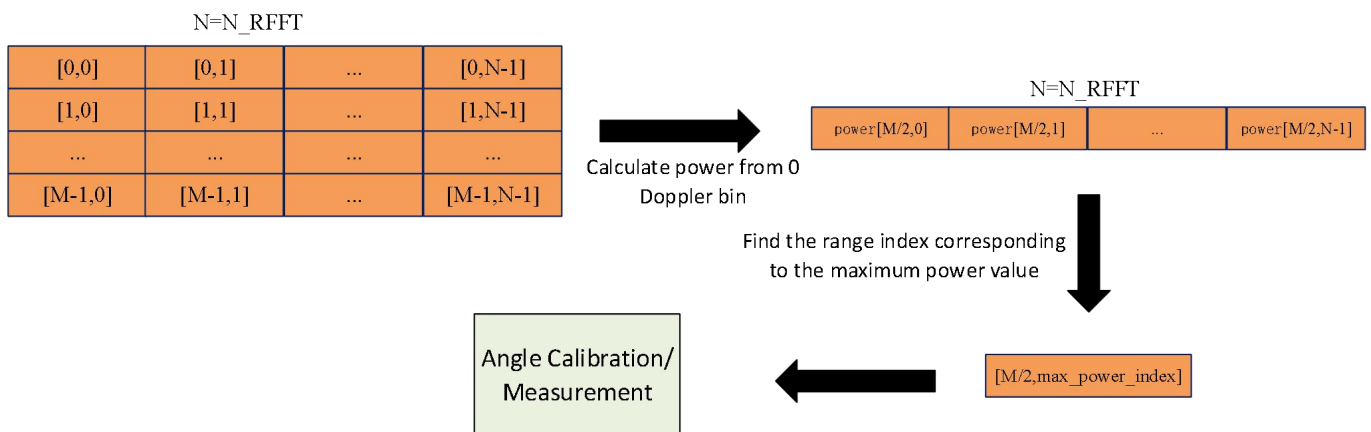


Figure 3-22 Schematic diagram of angle calibration/measurement data acquisition

3.3.3.2.3 Radar triggering function

Taking the radar triggering function with the wave-to-illuminate example, we verify whether the radar system is working properly from the application level. By calling the *mmw_point_cloud_process()* interface in *mmw_app_pointcloud.c*, we obtain the point cloud processing result and then determine whether there is point cloud within a specified distance range. The default distance range is configured as 1.5m.

3.3.3.2.4 Save calibration values

For RS6130, when saving calibration values to flash, the calibration value of Rx2 needs to be inverted. However, when saving calibration values to registers, no inversion is required. For RS6240, calibration values do not need to be saved to registers. When saving to flash, the calibration values can be directly saved. The format for saving calibration values in flash is with the real part first and the imaginary part second, arranged in the order of antennas. For specific code implementation, refer to the *mmw_angle_save_calib_data()* interface in the *ft_angle_est.c* file.

3.3.3.2.5 Obtain calibration values

For RS6130, when saving calibration values to flash, the Rx2 calibration value is inverted. Therefore, in order to obtain the correct calibration value, after reading the calibration values from flash, the Rx2 calibration value needs to be *inverted*. For specific code implementation, refer to the *mmw_angle_get_calib_data()* interface in the *ft_angle_est.c* file.

3.4 Requirements For Testing Environment

The functional test of the radar sensor requires an unobstructed environment space directly ahead, and no moving objects or personnel walking should occur during the test.

3.4.1 Environment selection

The selection of test environment space is referred to the schematic diagram below.

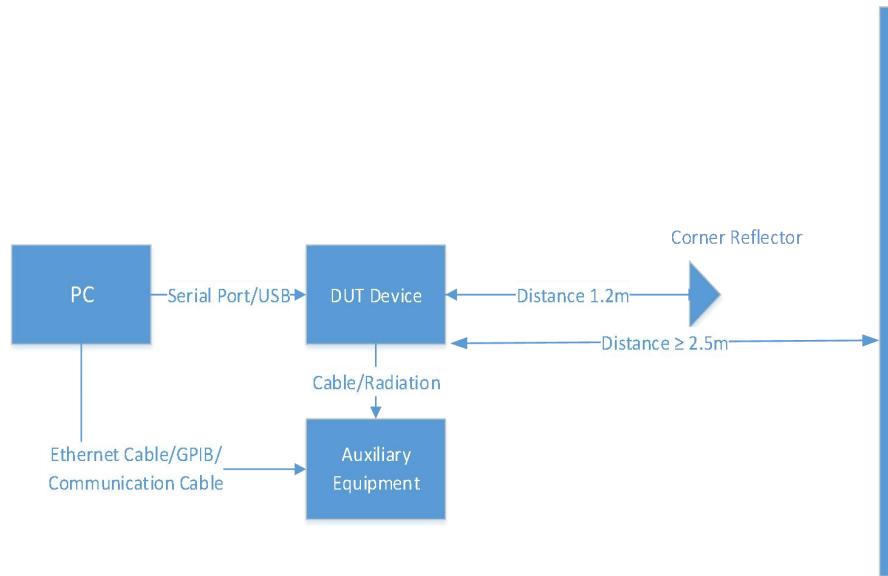


Figure 3-23 Schematic diagram of the environment

Environmental space requirements:

- Place the corner reflector $1.2\text{m} \pm 0.02\text{m}$ directly in front of the testing equipment. The area within 2.5m in front of the testing equipment needs to be kept clean, and areas beyond 2.5m can be non-metallic walls;
- Within the hemispherical area ahead, measured from the test equipment to an angle reflector positioned 1.2m away, no other strong reflective objects or moving targets are allowed. Additionally, beyond a distance of 1.2m ahead, there must be no right-angle objects, metal objects, or metal walls that resemble angle reflectors;
- Using a prototype for testing, the signal strength within a semi-circular range centered on the prototype and with a radius of 3 meters must be less than the retro-reflective power.
- The radar module should be positioned at a distance greater than 1.5m from both sides of the wall. The walls on both sides should not be made of metal, metal objects, or smooth mirrors, which are prone to reflecting millimeter-wave signals.

3.4.2 Environment setup

The setup of the test equipment and the corner reflector environment is shown in the figure below:

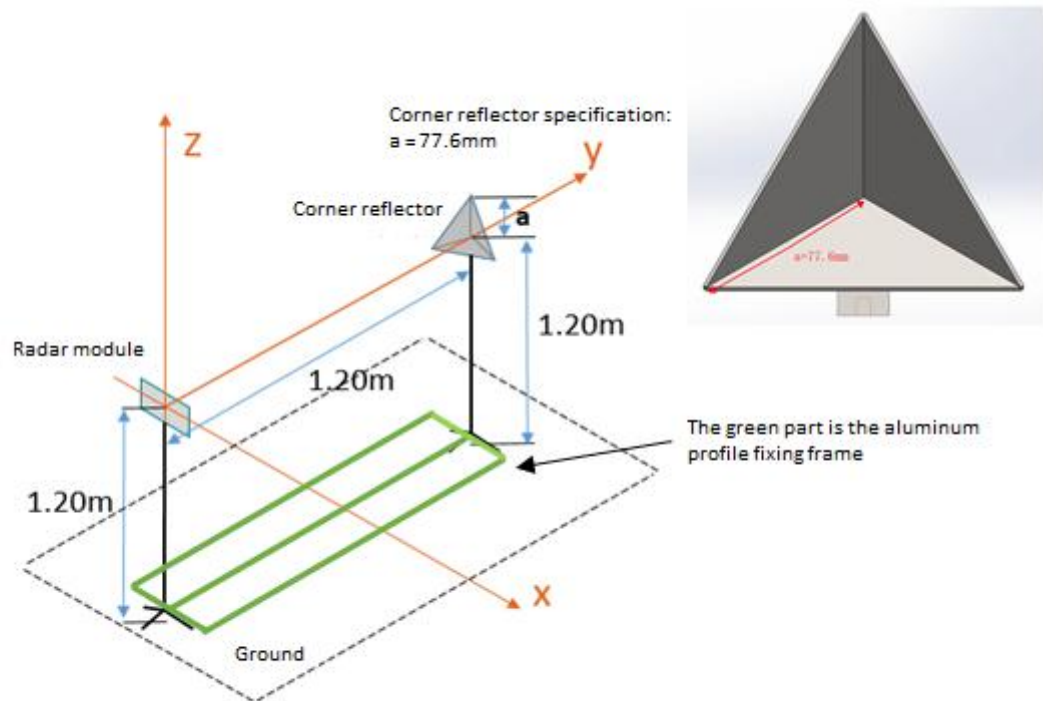


Figure 3-24 Setup of Test Environment

Requirements for setting up the testing environment:

- It is recommended that the specification for the angle reflector be set as $a = 77.6\text{mm}$;
- The test equipment and the corner reflector are positioned $1.2\text{m} \pm 0.02\text{m}$ above the ground, with the corner reflector situated $1.2\text{m} \pm 0.02\text{m}$ directly in front of and facing the test equipment;
- The testing equipment is perpendicular to the ground;
- The corner reflector with an open triangular cross-section is perpendicular to the ground, with the notch facing the radar. The center is at the same level as the test equipment (radar chip) (the radar must be within a horizontal angle of $0^\circ \pm 0.5^\circ$ and a pitch angle of $0^\circ \pm 0.5^\circ$ relative to the corner reflector).
- To ensure that the environment is not easily damaged, it is recommended to mount the radar module and the corner reflector on the same rigid fixture, such as a platform constructed from aluminum profiles. The support column for both the radar and the corner reflector is recommended to be made of bakelite, and the diameter of the corner reflector support column should be less than 2.5cm to avoid excessive reflection interference with the reception of the corner reflector signal.
- If multiple radars need to be set up for parallel testing, the test fixtures can be placed in parallel, but it is necessary to ensure that the distance between the radars is greater than 2 meters.
- If the actual environment cannot fully meet the above conditions, users may exercise discretion and verify the usability of the environment in accordance with section 3.4.3 "Test Environment Acceptance".

3.4.3 Test environment acceptance

Radar performance testing is sensitive to the environment, and users need to accept the testing environment according to the following methods.

- (1) **Using the development board (MRS6xxx-DEV) provided by our company**, collect 10 sets of ambient noise test data according to the instructions in Section 3.3.2.3 "Measure the background noise Level of the environment". When collecting ambient noise, please set the maximum trigger distance to 7m, the starting frequency point to 60.5GHz, and the distance resolution to 8cm.

- (2) Compare with the typical values in the table below. If the results are within a reasonable range, the environment is considered qualified. Otherwise, please refer to "Section 3.4.1 Environment Selection" for troubleshooting.

The data for each antenna of RS6130 must meet the following requirements:

Table 3-5 RS6130 R4 No-development solution reference value

Indicator	Minimum Value (dB)	Typical Value (dB)	Maximum Value (dB)
NoiseHighAvge	-74	-70	-66
NoiseLowAvge	Don't care	-80	Don't care
NoiseAvge	-73	-71	-69
PeakPwrAvge	-11	-9	-7
PeakSnrAvge	59	61	63

If user is using SDK2.0 as their software environment, the following items should be noted:

- Due to changes in the noise floor test for SDK2.0, the output values have also changed accordingly. Therefore, it is necessary to refer to the values in Table 3-3.
- Since the tool introduced in Section 3.3.2.3 "Measure the background noise Level of the environment" only supports the R4 No-development solution, SDK2.0 users need to obtain measurement results by calling the API.

Table 3-6 RS6130 SDK2.0 solution reference value

Indicator	Minimum Value (dB)	Typical Value (dB)	Maximum Value (dB)
NoiseHighAvge	-112	-108	-104
NoiseLowAvge	不卡控	-115	不卡控
NoiseAvge	-112	-110	-107
PeakPwrAvge	-42	-40	-38
PeakSnrAvge	64	68	73

The data for each antenna of RS6240 must meet the following requirements:

Table 3-7 RS240 SDK2.0 solution reference value

Indicator	Minimum Value (dB)	Typical Value (dB)	Maximum Value (dB)
NoiseHighAvge	TBD	TBD	TBD
NoiseLowAvge	TBD	TBD	TBD
NoiseAvge	TBD	TBD	TBD
PeakPwrAvge	TBD	TBD	TBD
PeakSnrAvge	TBD	TBD	TBD

3.4.4 Pre test environmental assurance

Before each batch test, it is necessary to ensure that the testing environment is in a normal state to guarantee the smooth progress of subsequent tests.

The following needs to be confirmed:

- Fixture: Check whether the model of the test fixture corresponds to the model of the module being tested, and whether the connecting wires and probes on the fixture are intact;
- Test 10 gold sample machines (the machines identified in Section 3.3.1.9 "Method for Determining Performance Test Indicators") to ensure that the testing environment is in a normal state.

The testing environment inspection process is as follows:

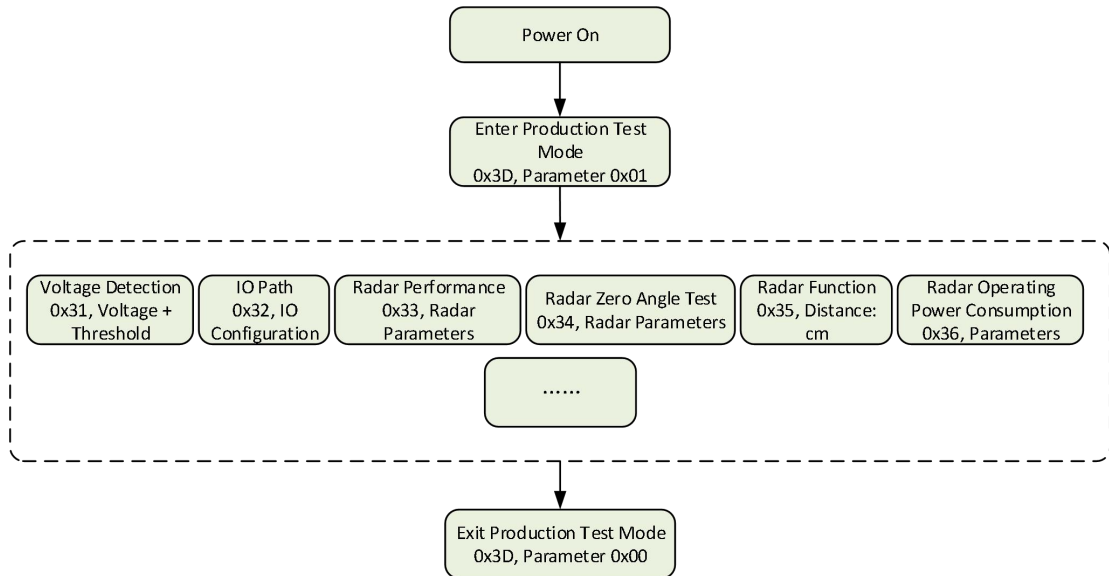


Figure 3-25 Test Environment Inspection Process

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